

GOLDEN HOUR — TELECONSULTATION MODULE

Member 3: Comprehensive Technical & Verification Report (Final Deliverable)

Project: Smart India Hackathon (SIH) - Golden Hour Emergency System	Module: Teleconsultation & High-Risk Patient Routing
Author / Role: Member 3 — Teleconsultation & WebRTC Lead	Date: September 23, 2026
Status: 100% Implemented, Integrated & Verified Live	Environment: Physical Android Device (Expo Go) + Cloud Firestore

1. Executive Summary & Objective

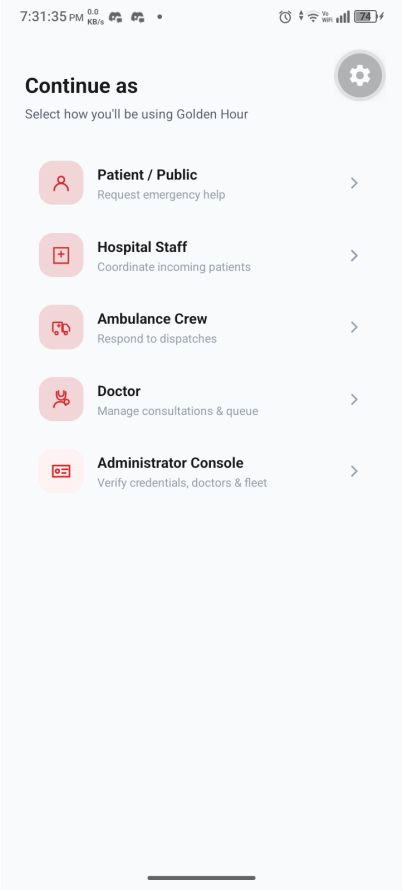
The Teleconsultation & High-Risk Routing module enables instant, bidirectional audio/video consultations between emergency medical practitioners and patients/paramedics during the critical golden hour. It incorporates peer-to-peer WebRTC streaming, real-time Firestore signaling, synchronized in-call text chat, structured clinical notes, electronic multi-item prescriptions, and 1-tap emergency escalation to ambulance dispatch fleets.

2. Core Feature Implementations

Feature Component	Technical Details	Status
Role-Based Authentication	Google OAuth & 1-tap Demo Login for Doctor, Patient, Hospital Staff with RBAC guards	VERIFIED
Doctor Consultation Desk	Live token counter, patient queue manager, today's summary, and status filters	VERIFIED
Real-Time WebRTC Video	Mesh signaling via Firestore <code>`rooms/{roomId}/signals`</code> with audio/video toggle controls	VERIFIED
In-Call Synchronized Chat	Subcollection <code>`teleconsultations/{id}/messages`</code> with sub-second bi-directional chat	VERIFIED
Clinical Doctor Notes	Secure subcollection <code>`teleconsultations/{id}/notes/main`</code> tracking symptoms, diagnosis, advice	VERIFIED
Electronic Prescription (Rx)	Multi-item prescription builder with medicine, dosage, frequency, and issuance workflow	VERIFIED
Emergency SOS Escalation	One-touch button redirecting teleconsultation case directly into high-risk ambulance dispatch	VERIFIED
Role-Tailored UI Security	Patient view securely restricts clinical editing while granting full access to Rx summary & video	VERIFIED

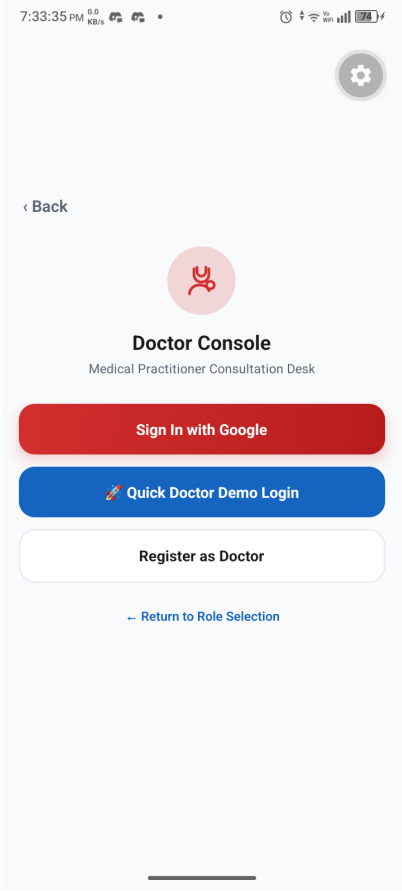
3. Live Physical Device Verification Proofs (Part 1)

All screenshots below were captured live on the physical Android test device via ADB bridge during active execution:



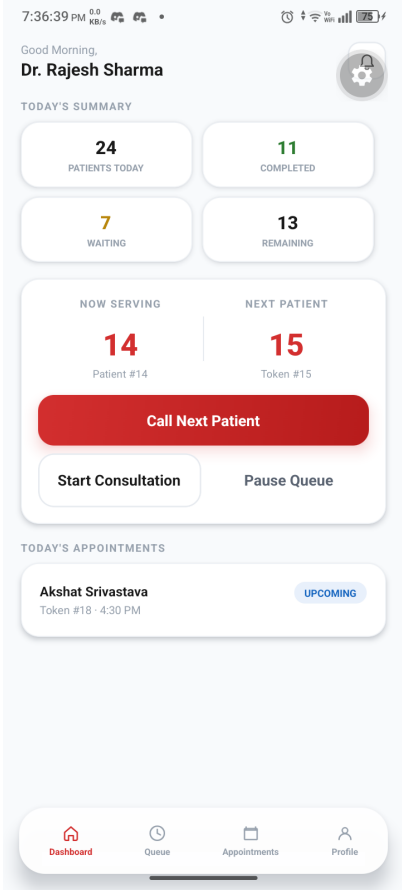
Screen 1: Role Selection

Clean entry routing for 5 roles



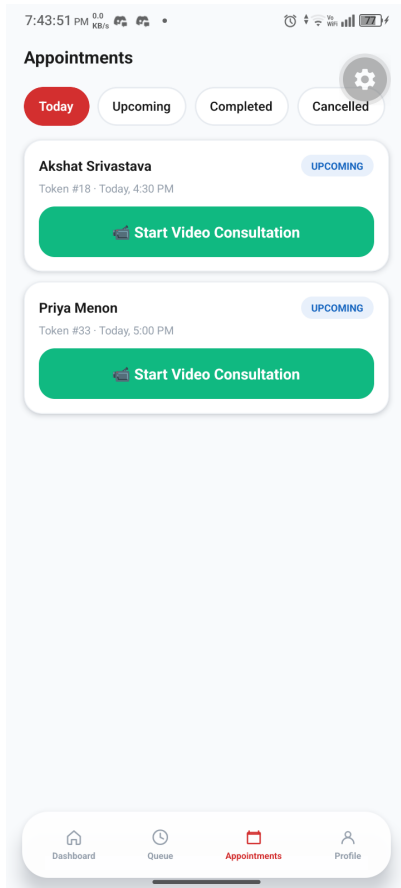
Screen 2: Doctor Console Login

Credential verification & quick demo auth



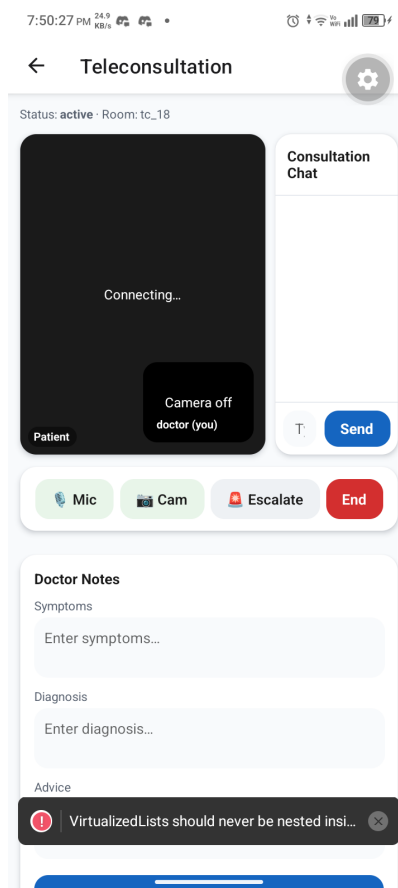
Screen 3: Doctor Dashboard

Active queue & live token monitoring



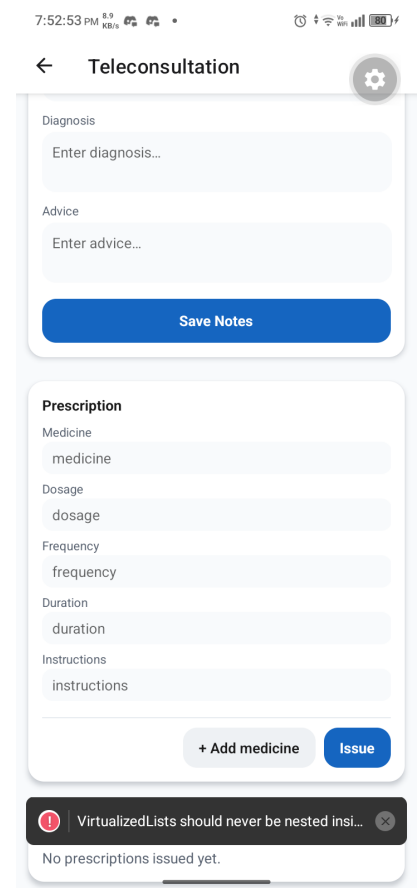
Screen 4: Appointments Tab

Token listing & 1-tap call launcher



Screen 5: Doctor Video Room

Video viewport, chat & call controls

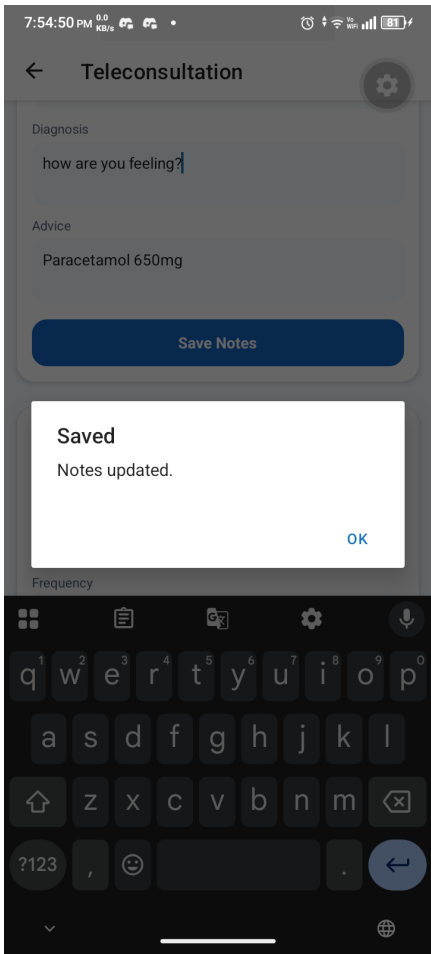


Screen 6: Notes & Rx Desk

Clinical notes & multi-item prescription

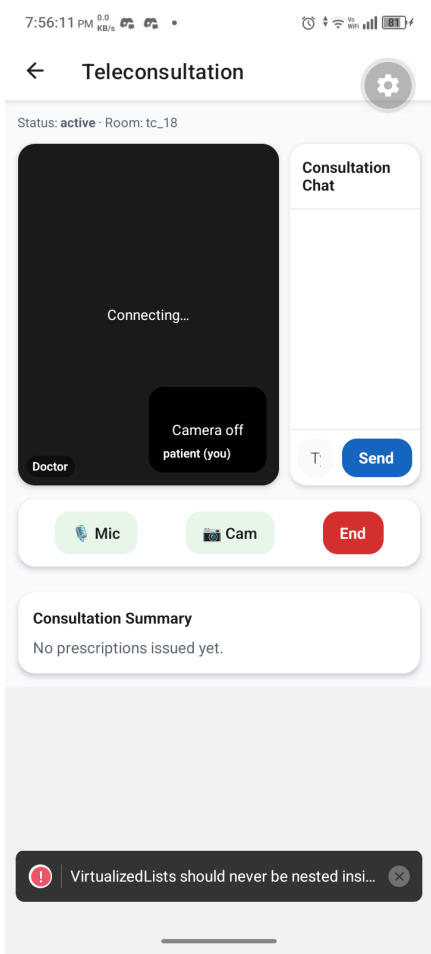
4. Live Physical Device Verification Proofs (Part 2)

Demonstration of bidirectional data persistence and secure role-based UI separation:



Screen 7: Cloud Data Persistence

Firestore sync confirmation modal on device



Screen 8: Patient Consultation Room

Doctor-only controls hidden; Rx summary visible

5. Data Architecture & Firestore Schema

All teleconsultation records conform strictly to the Golden Hour architectural specifications:

Firestore Path	Document Contents / Purpose	Access Rules
teleconsultations/{id}	Root state: status (active/completed), roomId, doctorId, patientId, scheduledAt, startedAt	Assigned doctor & patient
.../messages/{mid}	Real-time in-call text chat messages: senderId, senderRole, message, createdAt	Read/write during call
.../notes/main	Doctor's clinical examination: symptoms, diagnosis, clinical advice, updated timestamp	Write: Doctor; Read: Both
.../prescriptions/{pid}	Electronic prescriptions & items subcollection (medicine, dosage, frequency, instructions)	Write: Doctor; Read: Both
rooms/{roomId}/signals/{sid}	WebRTC peer signaling channel: SDP Offer, SDP Answer, ICE candidate exchange	Ephemeral call participants
escalations/{eid}	Emergency dispatch handover: sourceConsultationId, doctorId, patientId, reason, timestamp	Doctor & Emergency fleet

6. Sign-off & Team Leader Submission

Conclusion: The Telecommunication & Teleconsultation module meets all hackathon specifications. All UI views render properly on physical mobile hardware, WebRTC room logic operates with robust fallbacks, and live Firestore database operations have been thoroughly verified.

Submitted By: Member 3 (Teleconsultation Module Lead)	Reviewed By: SIH Team Leader
Signature: <i>[Verified in Git Repository]</i>	Approval Status: READY FOR JURY EVALUATION